

BCS102 – Fundamentals of Artificial Intelligence ASSIGNMENT

Course Assessment Details

Release date:	7/7/26
Due date:	28 July 2026 (Tue) by 5.00pm
Value:	40% of total assessment
Marks	Marked out of 100
Assessment Mode:	Individual/Group Assessment (Group of 2)
Length/Word Count:	Minimum 3000 words Maximum 3200 words

RATIONALE

This assignment is designed to allow students to demonstrate their understanding and ability to find supporting details in justifying the computing problem and research areas relevant to the chosen project/area.

Course Learning Outcomes

On completion of this assignment, students should be able to:

CL03: Examine the current scope, limitations and societal implications of artificial intelligence from existing literature (C4, PLO9, MQF4a).

1. ASSESSMENT QUESTION

A. ASSIGNMENT EVALUATION

These criteria will be used to evaluate your assignment submission:

- Python Notebook file is well commented and documented to guide readers to understand your assignment.
- Usage of Python libraries
- Suitable data loading, preparation
- An accurate Artificial Intelligence model is trained.
- Presentation of performance evaluation of the trained Artificial Intelligence model.

B. REQUIREMENTS

You are to load a dataset of your choice to Python Notebook and utilize suitable Python libraries to prepare and transform the data, then train a suitable Artificial Intelligence algorithm.

C. SUBMISSION

You are required to submit 2 items:

Item (1): Report that includes all the required tasks mentioned in the Marking Scheme Rubrics on page 4. Include the relevant screenshots.

You must submit this file to LMS (the provided link at LMS) before the submission time of the due date. This submission feature will automatically check for Turnitin's similarity score, therefore students must ensure that their writing is original. Penalty will be given for Turnitin similarity score above 15%; the affected task/s will be given zero (0).

Item (2): Properly commented and documented Python Notebook file.

You must submit this Python Notebook file to LMS (the provided link at LMS) before the submission time of the due date.

Case Study – Dataset

For this assignment, choose any ONE of the following domains:

- 1 Travel & Tourism
- 2 Healthcare <http://kaggle.com/datasets/prasad22/healthcare-dataset/data>
- 3 Logistics
- 4 Financial

Please use a suitable dataset of your choice from a chosen industry with an appropriate number of observations (more than 1000s, etc) for this assignment. List and describe briefly all the features (engineering and/or raw features) within this dataset. Explore the dataset (include visualization), prepare the data, propose to train using suitable Artificial Intelligence algorithm with appropriate setting, and propose to evaluate the performance of the trained algorithm based on suitable performance indicators, with ethical, social and future implications.

You should report how you collected, processed and analysed dataset. Discuss why you choose the AI algorithm, and ethical, social and future implications of your solution.

Dataset chosen: Healthcare <http://kaggle.com/datasets/prasad22/healthcare-dataset/data>

Total number of observation

All features/ column

Explore the dataset (include visualization)

prepare the data

Naive Bayes - propose to train using suitable Artificial Intelligence algorithm with appropriate setting, and propose to evaluate the performance of the trained algorithm based on suitable performance indicators, with ethical, social and future implications.

You should report how you collected, processed and analysed dataset. Discuss why you choose the AI algorithm, and ethical, social and future implications of your solution.

2. SUBMISSION GUIDELINES

3.1 Assignment Formatting

- File format: PDF or Microsoft Word.
- Font: Times New Roman or Arial, size 12.
- Line spacing: 1.5 and 1-inch margins on all sides
- Page numbering must be included.
- The formatting requirements may be adjusted only if the lecturer provides specific instructions or allowances.
- Any clarification regarding the assignment requirements should be directed to the principal lecturer of the subject.
- Students should keep a copy of all submitted assignments for their own records.

3.2 Assignment Cover Sheet & Marksheet

- Students must include the **Assignment Cover Sheet** containing the following details:
 - Full Name
 - Student ID
 - Course code: Course name
 - Lecturer's name
 - Assignment due date
 - Assignment Title
 - Word Count
- The **Marksheet/Marking Rubric** must be included as the final section of the assignment submission.

3.3 Submission Procedure

- All assignments must be submitted through **HLMS by 5.00 pm** on the specified due date.
- **Resubmission** is allowed in Turnitin.

Turnitin Resubmission Policy:

- Students may resubmit their assignment before the submission deadline.
- A maximum of three (3) submissions is allowed within a 24-hour period.
- Each new submission will replace the previous version of the assignment.
- After the third submission within 24 hours, Turnitin imposes a mandatory 24-hour waiting period before a new similarity report is generated.
- No additional time or deadline extension will be granted due to delays in obtaining a Turnitin similarity report after the submission deadline has passed.

3.4 Late Submission Policy

- A **penalty of 5 marks per day** will be imposed for late submissions. Sundays and public holidays will not be included in the penalty calculation.
- Students requiring an extension due to valid reasons must submit a **Late Submission Form** and obtain approval from the respective lecturer **at least three (3) days** before the original submission deadline.
- The approved Late Submission Form must be attached together with the assignment submission.
- An extension of up to one (1) week from the original due date may be granted at the lecturer's discretion.
- Any submission beyond the approved extension period will be considered a late submission and subjected to penalties.
- Assignments submitted **more than one (1) week** after the original due date (inclusive of weekends and public holidays) will not be accepted and will receive **ZERO (0) marks**.

3. ACADEMIC INTEGRITY AND REFERENCING

Honesty and Responsibility

Academic integrity is an important tenet for HELP University (the "University"). In pursuit of the highest standards of academic integrity, the University holds its students to the highest ethical standards defined by the Rules and Regulations section of the Academic Handbook. All students studying a HELP University programme are subjected to and are bound by the Student Academic Misconduct Rule to assure academic honesty. Students are required to sign a pledge on the assignment cover sheet before submitting your assignments.

- All sources, including textbooks, journal articles, websites, and AI-generated content, must be appropriately acknowledged and cited using **APA Style (7th Edition preferred)**.
- When in doubt, you can refer to the APA Style website for more information: <https://apastyle.apa.org/>
- Copying another student's work is strictly prohibited and may result in both students receiving zero (0) marks for the assignment.

- Failure to acknowledge any borrowed ideas, quotations, or materials may constitute academic misconduct and may result in penalties according to the University's Academic Integrity Policy.

4.1 Turnitin Similarity and AI Compliance

- The maximum allowable thresholds are as follows:
 - **Turnitin similarity score: ≤ 20%**
 - **AI-generated content score: ≤ 30%** (refer to section 4.2)
- Students are responsible for ensuring that their submissions comply with the stated thresholds before the submission deadline.
- Students are strongly advised to submit their work early to allow sufficient time for Turnitin report generation and resubmission, where applicable.
- Submissions exceeding the permitted Turnitin similarity threshold may be subject to mark deductions or an academic integrity review in accordance with University policy.
- Submissions with AI-generated content **exceeding 30%** will incur a penalty of **one (1) mark deduction for every additional 1% above** the permitted threshold.

4.2 Use Of AI Text Generators (AITG)

Terms and Conditions for using AITG

AITGs can be used on limited basis for assessments in this course.

Any use of AITGs must be properly acknowledged.

Any cases in which Turnitin detects more than **30% AI-generated content** in an assignment will be referred for review **when AI tools may be used on a limited basis**

- AI Text Generators (AITGs) may be used on a limited basis for brainstorming, generating ideas, and improving the structure of the assessment.
- Any use of AI-generated content must be properly acknowledged and cited.
- The use of AI tools without proper acknowledgment constitutes academic misconduct and may result in penalties as stated in the Student Academic Misconduct Rules.
- If concerns arise regarding AI-generated content in a submission, students may be required to explain their work, including their arguments, sources, drafts, and development process.
- Cases where Turnitin detects AI-generated content exceeding 30% may be reviewed to determine whether AI tools have been used appropriately and in accordance with the assessment requirements.

1. Utilizing AI generative tools should be seen to augment creativity, problem-solving, and productivity while maintaining the integrity of the subject matter and upholding academic and professional standards.

2. Students must demonstrate their own knowledge, skills, and understanding of the subject matter in their work.
3. Where an assignment requires ChatGPT to be cited, you must reference all the content from Generative AI tools that you include. Failure to reference externally sourced, non-original work can result in Academic misconduct.
4. Students are to apply **APA Style** for any AI content generated that is utilized and integrated into the final work. This acknowledgment is to be added to the **footnote** of the respective pages.

- **To cite AI-generated work that you did not edit or revise:**

- Name of AI Tool. (Year, Month Day you generated the content). Exact text of question or prompt you entered [AI-generated text/image/video, etc.]. Name of Company/Developer if different than name of AI tool. URL.
- Example: ChatGPT. (2023, June 3). "Steps of creating a running website using XAMPP?" [AI-generated text]. OpenAI. <https://chat.openai.com>.

- **To cite AI-generated work that you edited or revised:**

- Name of AI Tool & Your Last Name, First Initial. (Year, Month Day you generated the content). Exact text of question or prompt you entered [AI-generated text/image/video, etc.]. Name of Company/Developer if different than name of AI tool. URL.
- Example: ChatGPT & Chong, L.Y. (2023, July 19). How to setup cloud based CI/CD for a website deployment? [AI-generated text]. OpenAI. <https://chat.openai.com/c/9bb6771b-209b-4c8c-ac79-6a8a9f39604a>

- **General Format for In-Text Citations**

- Examples: To make fluffy basmati rice, "rinse the rice twice in cold water" (ChatGPT, 2023)
 - Cloud-based CI/CD requires the following steps to be applied (ChatGPT & Chong, L.Y., 2023)
5. Any violation of this policy will result in appropriate academic sanctions, which may include penalizing your grades, failing the course, and other disciplinary actions.

6. If students used AITG in their assessment, they must provide proper acknowledgements of its use in their work. They must describe the precise AI tools that were utilised, how they were applied, and the prompts that were employed. Their assignment should have a statement at the bottom with the following details:
[I acknowledge the use of ChatGPT by OpenAI to assist in generating ideas, improving grammar, and organising the content of this assignment.](#)
The prompts used include:
 - ["Explain the concept of cyber ethics in simple terms."](#)
 - ["Provide examples of strong and weak ties in social networks."](#)
 - ["Rewrite this paragraph in a more academic writing style."](#)

- “Suggest points for the conclusion section of my report.”

The output from these prompts was used to support brainstorming, improve sentence structure, and enhance the clarity and presentation of the assignment. All final editing, verification of information, and submission were completed by the student.

4. REQUIREMENTS

- To obtain a pass in the subject:
 - Student should pass **BOTH** Continuous Assessments (CA) and Final Assessments (FA)
 - Student must attempt **ALL** coursework components in order to pass the course.

5. MARKING RUBRIC

BCS102 Marking Scheme Rubrics Assignment – Semester 2, 2026

CLO assessed	Topic			Marks awarded
CLO3	Data Exploration Python			
	Level C – Minimum Data Exploration and Data Shape Analysis; dataset dimension (0-3)	Level B – Considerable Data Exploration and Data Shape Analysis: dataset dimension, features’ data types (4-7)	Level A - Comprehensive Data Exploration and Data Shape Analysis: dataset dimension, features’ data types, feature renames, data type conversions (8-10)	
CLO3	Data Visualization using Python			
	Level C – Graphs provided with brief explanation (0-3)	Level B – Graphs provided with considerable explanation (4-7)	Level A - Graphs provided with comprehensive explanation i.e., presumed distribution shape, outliers, etc (8-10)	
CLO3	Data Preparation using Python			
	Level C – Minimum data preparation: correlation, feature extraction (0-3)	Level B - Considerable data preparation: correlation, feature extraction, cleaning (4-7)	Level A - Comprehensive data preparation i.e., correlation, feature extraction, cleaning, transformation, scaling, etc (8-10)	
CLO3	Data Splitting using Python			
	Level C – Minimum explanation on processes and method selection for data splitting (0-3)	Level B - Considerable explanation on processes and method selection for data splitting (4-7)	Level A - Comprehensive explanation on processes and method selection for data splitting (8-10)	
CLO3	AI Algorithm on the Training Set using Python			
	Level C – Implement the chosen Artificial Intelligence algorithm using the correct libraries on the training set without justification (0-8)	Level B – Implement the chosen Artificial Intelligence algorithm using the correct libraries on the training set with considerable justification and tuning of hyperparameters (9-17)	Level A – Implement the chosen Artificial Intelligence algorithm using the correct libraries on the training set with comprehensive justification and tuning of hyperparameters (18- 25)	
	Evaluate the Trained AI Algorithm on the Test Set using Python			

CLO3	Level C – Evaluate the trained algorithm on the test set with minimum discussion of suitable performance (0-8)	Level B – Evaluate the trained algorithm on the test set with considerable discussion of suitable performance indicators and ethical and future implications (9-17)	Level A – Evaluate the chosen algorithm on the test set with comprehensive discussion of suitable performance indicators and ethical and future implications (18-25)	
Python Notebook File				
CL03	Level C - Insufficient comments and documentation. (0-3)	Level B – Minimum comments and documentation. (4-7)	Level A - Comprehensive comments and documentation. (8-10)	
TOTAL MARKS				



Assignment No.: _____

Assignment Cover Sheet

Student Information (For group assignment, please state names of all members)		Grade/Marks
Name	ID	

Module/Subject Information		Office Acknowledgement
Module/Subject Code		
Module/Subject Name		
Lecturer/Tutor/Facilitator		
Due Date		
Assignment Title/Topic		
Intake (where applicable)		
Word Count		Date/Time

Declaration

- ☐ I/We have read and understood the Programme Handbook that explains on **plagiarism**, and I/we testify that, unless otherwise acknowledged, the work submitted herein is entirely my/our own.

- ☐ I/We declare that no part of this assignment has been written for me/us by any other person(s) except where such collaboration has been authorized by the lecturer concerned.
- ☐ I/We authorize the University to test any work submitted by me/us, using text comparison software, for instances of plagiarism. I/We understand this will involve the University or its contractors copying my/our work and storing it on a database to be used in future to test work submitted by others.

Note: 1) The attachment of this statement on any electronically submitted assignments will be deemed to have the same authority as a signed statement.

2) The Group Leader signs the declaration on behalf of all members.

Signature:	Date:
E-mail:	

Feedback/Comments*	
Main Strengths	
Main Weaknesses	
Suggestions for improvement	
	Student acknowledge feedback/comments
Grader's signature	Student's signature:
Date:	Date:

Note:

- 1) A soft and hard copy of the assignment shall be submitted.
- 2) The signed copy of the assignment cover sheet shall be retained by the marker.
- 3) If the Turnitin report is required, students have to submit it with the assignment. However, departments may allow students up to **THREE** (3) working days after submission of the assignment to submit the Turnitin report. The assignment shall only be marked upon the submission of the Turnitin report.

*Use additional sheets if required

REFERENCES (SAMPLE)

- Hung, H., Liu, I., Liang, C., & Su, Y. (2020). Applying Educational Data Mining to Explore Students' Learning Patterns in the Flipped Learning Approach for Coding Education. *Symmetry*, 12(2), 213. <https://doi.org/10.3390/sym12020213>
- Surbakti, H. (2015). Integrating Knowledge Management and Business Intelligence Processes for Empowering Government Business Organizations. *International Journal Of Computer Applications*, 114(5), 36-43. <https://doi.org/10.5120/19976-1874>
- Surbakti, H., & Ta'a, A. (2019). Cognitive Approach Using SFL Theory in Capturing Tacit Knowledge in Business Intelligence. 2019 International Conference On Data And Software Engineering (Icodse). <https://doi.org/10.1109/icodse48700.2019.9092615>
- Vermeulen. (2012). Free gathering and movement of evidence in criminal matters in the EU. Thinking beyond borders, striving for balance, in search of coherence. Maklu.
- What is a Data Warehouse? | Key Concepts | Amazon Web Services. Amazon Web Services, Inc. (2021). Retrieved 17 January 2021, from <https://aws.amazon.com/data-warehouse/>.

In-text Citation Sample:

You have many different operational applications, but right now all your reporting and analytics comes directly out of those applications (Hung et al., 2020). You decide that you need a data warehouse to support integrated reporting that will combine data from some or many of those source applications (Surbakti & Ta'a, 2019). One thing that you completely noticed is your data will be copied from the source systems and applications to the data warehouse, but also retained in those original sources for operational purposes. Vermeulen (2012) suggests to support your organization's analytics, you are going to use several different approaches in concert with one another. Specifically, you will be building a data warehouse; also building a data lake; and building data virtualization capabilities ("What is a Data Warehouse? | Key Concepts | Amazon Web Services", 2021).

Notes:

- *The sample above include books, journal papers, conference papers, proceedings, online references.*
- *ONLY refer papers, journals, and books for the last five years.*
- *Electronic resources (blogs, websites, electronic articles, etc) are NOT allowed.*
- *Order by author name (ascending order).*
- *MUST follow the APA 7th referencing style.*